COMPARISON OF DEPTH AND OF THE DEPTH-DIMENSION SUM BETWEEN A LOCAL RING AND ITS IDEAL-ADIC COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The behaviour of depth under a flat local homomorphism, together with the formal fibre formula. Invoked inside the counted proof at source lines 2585-2593 and attributed to a lemma of another chapter (a cross-chapter pointer). Splits the depth of the completion into the depth of the original module and the depth of the formal fibre.
- The Cohen-Macaulay property of the formal fibres of a Noetherian ring with a dualizing complex, and the equality of dimensions of a local ring and its completion. Both are invoked inside the counted proof (source lines 2594-2601 and 2617-2620) as cross-chapter pointers, the first with a reference to a further chapter of more algebra. Replaces the depth of the formal fibre by its dimension and identifies the two dimension terms appearing in the displayed chain of equalities.

Source reference. Original reference algebra-lemma-apply-grothendieck-module: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2593–2593. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-apply-grothendieck-module, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dualizing-lemma-dualizing-gorenstein-formal-fibres: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2596–2596. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-dualizing-gorenstein-formal-fibres, which belongs to another chapter of the

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Source reference. Original reference more-algebra-section-properties-formal-fibres: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2598–2598. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-properties-formal-fibres, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-completion-dimension: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 834–834. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-completion-dimension, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring which has a dualizing complex. Let $I \subset A$ be an ideal. Let M be a finite A-module. Let A', M' be the I-adic completions of A, M. Let $\mathfrak{p}' \subset \mathfrak{q}'$ be prime ideals of A' with $\mathfrak{q}' \in V(IA')$ lying over $\mathfrak{p} \subset \mathfrak{q}$ in A. Then*

$$depth_{A_{\mathfrak{p}'}}(M'_{\mathfrak{p}'}) \geq depth_{A_{\mathfrak{p}}}(M_{\mathfrak{p}})$$

and

$$depth_{A_{\mathfrak{p}'}}(M'_{\mathfrak{p}'}) + \dim((A'/\mathfrak{p}')_{\mathfrak{q}'}) = depth_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) + \dim((A/\mathfrak{p})_{\mathfrak{q}})$$

Proof. We have

$$\operatorname{depth}(M'_{\mathfrak{p}'}) = \operatorname{depth}(M_{\mathfrak{p}}) + \operatorname{depth}(A'_{\mathfrak{p}'}/\mathfrak{p}A'_{\mathfrak{p}'}) \geq \operatorname{depth}(M_{\mathfrak{p}})$$

by flatness of $A \to A'$, see Algebra, Lemma algebra-lemma-apply-grothendieck-module. Since the fibres of $A \to A'$ are Cohen-Macaulay (Dualizing Complexes, Lemma dualizing-lemma-dualizing-gorenstein-formal-fibres and More on Algebra, Section more-algebra-section-properties-formal-fibres) we see that $\operatorname{depth}(A'_{\mathfrak{p}'}/\mathfrak{p}A'_{\mathfrak{p}'}) = \dim(A'_{\mathfrak{p}'}/\mathfrak{p}A'_{\mathfrak{p}'})$. Thus we obtain

$$\begin{aligned}
\operatorname{depth}(M'_{\mathfrak{p}'}) + \dim((A'/\mathfrak{p}')_{\mathfrak{q}'}) &= \operatorname{depth}(M_{\mathfrak{p}}) + \dim(A'_{\mathfrak{p}'}/\mathfrak{p}A'_{\mathfrak{p}'}) + \dim((A'/\mathfrak{p}')_{\mathfrak{q}'}) \\
&= \operatorname{depth}(M_{\mathfrak{p}}) + \dim((A'/\mathfrak{p}A')_{\mathfrak{q}'}) \\
&= \operatorname{depth}(M_{\mathfrak{p}}) + \dim((A/\mathfrak{p})_{\mathfrak{q}})
\end{aligned}$$

Second equality because A' is catenary and third equality by More on Algebra, Lemma more-algebra-lemma-completion-dimension as $(A/\mathfrak{p})_{\mathfrak{q}}$ and $(A'/\mathfrak{p}A')_{\mathfrak{q}'}$ have the same I-adic completions. □